

Deployment Report

PrepMate – AI Interview & Project Viva Simulator

1. Introduction

This report summarizes the deployment of the PrepMate web application developed during the internship project. The objective was to make the application publicly accessible while ensuring stable communication between the frontend, backend, AI services, and database.

2. Deployment Platforms

Component	Platform
Frontend	Cloudflare Pages
Backend	Render
Version Control	GitHub
AI Service	Google Gemini API
Database	SQLite

3. Deployment Process

1. Source code was maintained in GitHub.
2. FastAPI backend was deployed on Render.
3. Environment variables were configured securely on Render.
4. Swagger documentation was verified through the /docs endpoint.
5. Static frontend was deployed using Cloudflare Pages connected to GitHub.
6. Frontend configuration was updated to communicate with the deployed Render backend.
7. End-to-end testing was performed after deployment.

4. Deployment Verification

The following functionality was verified after deployment:

- Home page loaded successfully.
- Technical Interview module functioned correctly.
- Project Viva module functioned correctly.
- Resume/report upload worked successfully.
- AI-generated responses were returned correctly.
- FastAPI Swagger documentation was accessible.
- Frontend and backend communicated successfully over HTTPS.

5. Challenges Faced

During deployment several issues were encountered, including Python version compatibility on Render, package installation errors, frontend configuration updates for production, static asset path corrections, and initial deployment issues with an alternative hosting platform. These were resolved by updating the deployment configuration, correcting resource paths, and deploying the frontend through Cloudflare Pages.

6. Conclusion

The PrepMate application was successfully deployed using Cloudflare Pages for the frontend and Render for the FastAPI backend. All major features were validated after deployment, confirming

that the application is operational and suitable for demonstration and further development.